

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Branch: AI&DS and Allied branches Semester :V Subject Code & Name: BTAIC501 Computer Network and Cloud Computing Max Marks: 60 Date: 01/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the various uses and applications of computer networks in different domains (e.g., business, education, healthcare)	2	6
B)	Compare and contrast different types of computer networks (e.g., LAN, WAN, MAN, PAN)	3	6
C)	Which of the OSI layers handle each of the following functions. a. Compression & encryption of data b. Determine the route through subnet to use. c. Provide synchronization. d. Flow control between adjacent machines physically connected by a communication channel. e. Reliable data transfer from source to destination independent of subnet	3	6
Q.2	Solve Any Two of the following.		12
A)	Explain the concept of error detection and correction in the data link layer. Discuss common techniques such as parity, checksums.	2	6
B)	Explain distance-vector routing algorithms used in the network layer	2	6
C)	Discuss techniques for congestion control, quality of service (QoS), and traffic shaping.	2	6
Q. 3	Solve Any Two of the following.		12
A)	Discuss the services provided by the transport layer to applications.	2	6
B)	Compare TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)	3	6
C)	Explain the functions of SMTP (Simple Mail Transfer Protocol) and POP/IMAP (Post Office Protocol / Internet Message Access Protocol).	2	6

Q.4	Solve Any Two of the following.		12
A)	Explain how these technologies contribute to the flexibility and scalability of cloud services.	2	6
B)	List and describe key enabling technologies used in cloud computing (e.g., virtualization, containerization).	2	6
C)	Identify potential risks associated with cloud adoption (e.g., security, data privacy, vendor lock-in).	2	6
Q. 5	Solve Any Two of the following.		12
A)	Discuss how EC2 instances are provisioned, managed, and scaled within the AWS cloud.	3	6
B)	Explain the concept of virtualization in the context of cloud computing.	2	6
C)	Discuss the functionalities typically offered by CaaS providers (e.g., telephony, messaging, conferencing)	2	6
	*** End ***		